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Unit 2 Project Submission

Exercise 1: Summarize Common APT Techniques and Procedures

BioGenX software packages passed an incorrect or improperly verified digital signature, which resulted in malware being introduced into the system. As the Chief Information Security Officer, it was important to further investigate and analyze the overall network activity. The investigation revealed repeated outbound connections to unfamiliar external domains, unencrypted HTTP communication, and disguised executable files. These findings strongly indicate that Advanced Persistent Threat (APT) actors may be present within the system, using multiple tactics to maintain access, evade detection, and achieve long-term objectives. As a cybersecurity professional, it is important to be aware of additional APT tactics beyond initial access methods. Two important APT tactics include Command and Control (C2) communication and persistence mechanisms.

Advanced Persistent Threat Tactic 1: Command and Control (C2) Communication

After initial access has been achieved, Command and Control communication allows attackers to remotely control compromised systems. Often referred to as C2 or C&C, this tactic involves covert communication channels between devices within the victim organization and infrastructure controlled by the attacker. Command and Control consists of hidden communication paths that allow attackers to send commands, receive system information, and maintain access to compromised environments (Gimmick, 2022).

There are multiple Command and Control techniques, but APT actors commonly disguise C2 traffic within legitimate protocols such as HTTP, HTTPS, DNS, encrypted traffic, or encoded data. This tactic was observed during the previous investigation, where repeated outbound connections to unfamiliar external IP addresses and domains were identified. Through C2 channels, attackers can issue commands, collect data, and prepare for data exfiltration while remaining connected to the environment.

Detecting Command and Control activity can be challenging because this traffic blends into normal network communication. As a result, attackers can remain persistent for long periods without triggering alerts. Security teams must focus on continuous monitoring and filtering of outbound traffic, inspecting traffic patterns, identifying beaconing behavior, and correlating data from multiple sources to detect C2 activity (Gimmick, 2022). While APT actors can operate quietly, implementing proper monitoring and mitigation controls significantly reduces the risk of long-term compromise.

Advanced Persistent Threat Tactic 2: Persistence Mechanisms

Another tactic commonly used by Advanced Persistent Threat (APT) actors is persistence mechanisms. Persistence is used to ensure continued access to a compromised system even after system reboots, user logouts, or partial remediation efforts. Once initial access is achieved, APT actors often install malware that creates scheduled tasks, startup scripts, registry modifications,

or malicious services. These mechanisms allow the malware to automatically re-execute, helping attackers maintain long-term presence while they observe the environment and plan their next actions (Shearer, 2023).

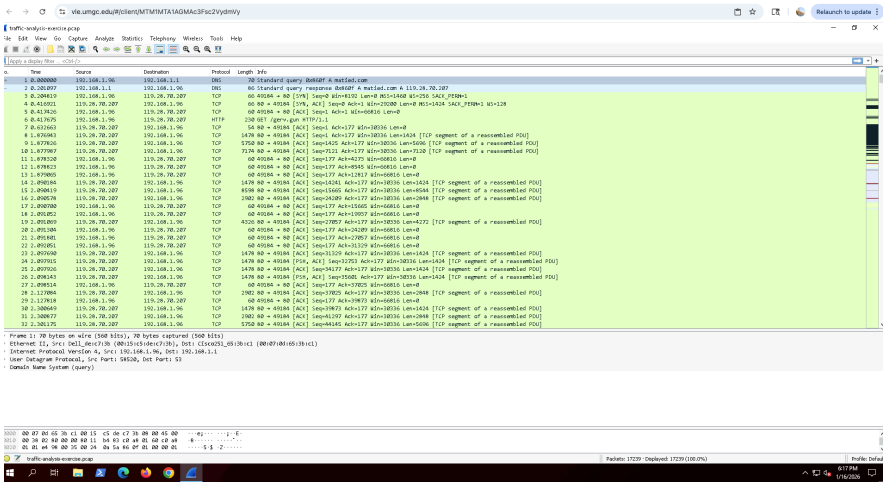
Persistence enables APT actors to remain hidden while gradually expanding their access within the organization. This long-term presence allows attackers to identify valuable assets, exploit additional vulnerabilities, and move laterally through the network. Once hidden backdoors are established, attackers can quietly maintain access, collect sensitive information, or prepare for data exfiltration and further malware deployment without immediate detection.

To reduce the risk of persistence-based attacks, BioGenX must implement strong mitigation strategies to protect organizational data and control system access. These strategies include managing vendor access, blocking unauthorized file writes, enforcing the principle of least privilege, using standard user accounts, implementing identity threat detection and response, monitoring and managing privileged access, preventing unwanted programs from running, and automating password management (Shearer, 2023). These mitigation techniques are critical for limiting an attacker's ability to maintain long-term access and reducing the overall impact of APT activity.

Overall, these APT tactic highlight how attackers move beyond initial compromise to achieve long-term control of targeted systems. Command and Control communication enables remote management and data exfiltration, while persistence mechanisms ensure attackers retain access over time. Understanding these techniques allows cybersecurity professionals to better detect advanced threats, identify root causes of compromise, and implement stronger monitoring, endpoint security, and automated validation controls to prevent future APT attacks.

Exercise 2: Use Network Security Tools to Detect Common APT Tactics

Part1: Wireshark Scan Results



I analyzed the PCAP file, *traffic-analysis-exercise.pcap,* using Wireshark to identify suspicious network activity related to Advanced Persistent Threats. After reviewing the captured traffic, I

observed DNS queries and repeated outbound connections from an internal system to an unfamiliar external domain, followed by HTTP communication. These findings indicate suspicious behavior that may be associated with APT tactics.

Part 2: PCAP Vulnerability Results Type

The PCAP analysis showed that the internal system made DNS queries and repeatedly communicated with an unfamiliar external domain using HTTP. HTTP traffic is commonly allowed on networks, which allows APT attackers to hide in plain sight. The attacker was able to disguise malicious communication as normal web traffic without being detected.

The vulnerability exploited in this case was weak network communication controls, specifically the use of unencrypted HTTP traffic and the lack of monitoring for outbound connections. Advanced Persistent Threat actors commonly take advantage of the HTTP and DNS protocols for command-and-control communication, which allows them to remain undetected while maintaining access to compromised systems. According to DNSFilter, unusual outbound DNS traffic and unencrypted communications are common indicators of malicious activity. MITRE ATT&CK also explains that adversaries frequently use DNS-based techniques to blend into normal network traffic and avoid detection(MITRE, 2025). Strengthening outbound traffic monitoring and enforcing secure communication protocols would help reduce the risk of similar APT activity.

Exercise 3: Examine Network Traffic Analysis and Monitoring Output to Confirm APTs

During the Wireshark file analysis, several traffic patterns were identified as unusual and concerning for the BioGenX system. One of the first anomalies observed was repeated DNS queries from an internal system to unfamiliar external domains. These findings strongly indicate an attempt to communicate with an external server that is not recognized as part of normal business activity. This behavior is unusual for typical user activity and may suggest that the system has been compromised and is attempting to maintain external communication. Repeated DNS activity of this nature is commonly associated with malware-controlled systems, where communication patterns are intentional rather than random. According to Halcyon, attackers often abuse DNS traffic because it is widely trusted and less closely inspected than other protocols, allowing malicious communication to bypass firewalls and intrusion detection systems more easily.

The traffic analysis also showed repeated outbound connections using HTTP rather than encrypted HTTPS. HTTP traffic is commonly allowed on networks, which allows malicious communication to disguise itself as normal web activity. In this case, the HTTP request appeared consistent and persistent, which suggests automated communication rather than user-initiated browsing. This pattern is often associated with command-and-control activity used by Advanced Persistent Threat actors. Without encryption, this type of traffic increases risk by allowing APT

actors to blend malicious communication into trusted network traffic without triggering immediate security alerts.

Another indicator of potential APT activity was the ongoing nature of communication. Instead of single connection or short session, the traffic identified repeated interactions with the same external destinations over time. This persistence aligns with how APT actors maintain long term access to compromised systems while voiding detections. APT actors commonly rely on sustained, low-volume communication to remain hidden while continuing to receive commands and transmit data.

When analyzed collectively, these traffic patterns demonstrate behavior that aligns with Advanced Persistent Threat characteristics rather than short-lived or opportunistic attacks. The consistent timing, repeated destinations, and protocol choice indicate intentional and structured communication. This behavior aligns with known command-and-control and persistence techniques documented in the MITRE ATT&CK framework, where adversaries abuse common protocols to maintain long-term access while avoiding detection (MITRE ATT&CK, 2025).

Overall, the combination of repeated DNS queries, persistent outbound HTTP traffic, and communication with unfamiliar external domains represents abnormal network behavior. These anomalies are strong indicators of Advanced Persistent Threat activity and demonstrate how attackers can remain hidden by using common network protocols to maintain control and access within a compromised environment.

Exercise 4: Validate Root Cause of Vulnerabilities Allowing for APTs

BioGenX's network was compromised due to an incorrectly verified digital signature, which shows that the issue extended beyond a single software download. The organization relied mainly on manual processes to verify software before installation. Depending only on manual verification increased the risk of human error and delayed the detection of malicious activity. To reduce this risk, automated controls are critical for preventing improperly signed software from executing within the network. Implementing automated verification helps block malicious software before attackers are able to gain access to internal systems.

Manual verification processes should only be used in limited circumstances, such as in-person identity verification, multi-factor authentication (MFA), or controlled remote verification scenarios. This approach is acceptable when trained human experts are responsible for confirming the authenticity of user-submitted identity documents and ensuring they align with the individual submitting them (CAF, 2024).

At BioGenX, relying on manual verification alone made the system more vulnerable to malicious software bypassing security checks and establishing a foothold within the environment. Without enforced automated identity and software verification controls, Advanced Persistent Threat actors can take advantage of these gaps to maintain access, communicate externally, and operate undetected within the network. This weakness allows attackers to move beyond initial compromise and establish long-term persistence within the network.

For incident prevention, BioGenX should implement automated software verification policies. Automated digital signature validation ensures that software is checked in real time before installation or execution. In many environments, automated security control assessments are integrated into platforms such as Security Information and Event Management (SIEM), Security Orchestration, Automation, and Response (SOAR), and vulnerability management systems (CyberProof Research Team, 2025).

These platforms are designed to continuously monitor security controls, alert security teams to activity that does not align with compliance standards, and provide evidence for audit and reporting purposes. With automated enforcement in place, files that are unsigned, altered, or issued by untrusted publishers can be automatically blocked or flagged for investigation. This approach reduces reliance on manual review and ensures consistent enforcement of security standards. Without automation, organizations may experience delayed certifications, increased risk exposure, and added strain on security teams (CyberProof Research Team, 2025).Manual verification processes also contribute to malware evasion, which is a common technique used by Advanced Persistent Threat actors. APT-related malware is often designed to appear legitimate by using trusted filenames, modified digital signatures, or seemingly valid publishers to avoid suspicion during manual review. When verification relies on human judgment rather than continuous automated validation, these evasion techniques can bypass detection. By exploiting gaps in manual verification, APT actors are able to maintain persistence, avoid security controls, and remain active within the network for extended periods of time. Implementing automated verification directly disrupts these evasion strategies by continuously validating software integrity and preventing unauthorized execution.

In addition, automated controls can be integrated into endpoint security tools such as application whitelisting, antivirus, or endpoint detection and response (EDR) solutions. These tools continuously validate software integrity and prevent unauthorized applications from running. According to Palo Alto Networks, combining machine learning, behavioral analytics, and threat intelligence allows endpoint solutions to detect and neutralize advanced malware, ransomware, and zero-day exploits that may evade traditional defenses. By enforcing automated verification policies rather than relying on manual verification, BioGenX can significantly reduce the risk of malicious software execution and limit the effectiveness of APT tactics.

Exercise 5: Reflect on APT Tactics

Have you or someone you know experienced a phishing attack? Do you believe that the experience meets the requirements of a focused APT attack? Justify your response using the resources in this unit.

While working at Chinese Laundry as an E-commerce Merchandising Assistant, I experienced a phishing attack that demonstrated how cybersecurity threats can impact employees across all departments. In my role, I was responsible for managing the company's wholesale website and had access to sensitive business and network data that required confidentiality, integrity, and availability. The position was fast-paced and involved frequent communication with brand managers, the IT team, and the accounting department. On a daily basis, I received emails from

senior-level employees requesting access to documents or system information, which made email communication a critical part of my responsibilities. Because of this risk, the company conducted annual cybersecurity awareness and phishing training for employees.

During this time, I received an email that appeared to come from a high-level brand manager requesting information that seemed unusual. At first glance, the email looked legitimate and closely matched the brand manager's actual email address. However, upon closer inspection, I noticed a small difference in the email address involving punctuation. This minor detail could have easily been overlooked without proper training. To confirm my concerns, I went directly to the brand manager's office to verify the request. She confirmed that she had not sent the email and was unaware that her identity was being impersonated.

After identifying the phishing attempt, I immediately reported the incident to the IT security team and did not respond to the email. The IT team quickly sent a company-wide alert warning employees about the phishing attempt and advised staff to remain cautious of similar emails. In addition, employees were required to complete updated phishing awareness training. This early detection and response helped reduce the risk of the attacker gaining access to sensitive systems or data and limited the potential impact of the incident.

This experience highlighted how cybersecurity is not limited to IT departments but is a shared responsibility across an organization. Prior to taking this course, I was not fully aware of the level of coordination, monitoring, and training required to protect organizational networks. This incident reinforced the importance of user awareness, early detection, and proper reporting in preventing Advanced Persistent Threat actors from exploiting human vulnerabilities to gain access to systems.

References:

- Cynet. (2026, January 13). *What are Advanced Persistent Threats (APT) & 5 defensive measures*. Cynet Unified, AI-Powered Security Platform for MSPs & SMEs. https://www.cynet.com/advanced-persistent-threat-apt-attacks/
- Grimmick, R. (2022, August 8). What is C2? Command and Control Infrastructure Explained. *varonis*. https://www.varonis.com/blog/what-is-c2
- *What is persistence in cybersecurity and how do you stop an advanced persistent threat (APT)?* (2023, June 6). Beyondtrust. https://www.beyondtrust.com/blog/entry/what-is-persistence-in-cybersecurity
- Command and Control, Tactic TA0011 - Enterprise | MITRE ATT&CK®. (n.d.). https://attack.mitre.org/tactics/TA0011/
- DNSFilter, Inc. (n.d.). *Command and Control Attack | Glossary | DNSFilter*. DNSFilter. https://www.dnsfilter.com/glossary/command-and-control-attack
- *What is DNS? - A DNS Security Primer*. (n.d.). https://www.centerforcybersecuritypolicy.org/insights-and-research/what-is-dns---a-dns-security-primer
- *What is a digital signature? Definition, security & more | ProofPoint US*. (2024, December 12). Proofpoint. https://www.proofpoint.com/us/threat-reference/digital-signature
- Team, C. R., & Team, C. R. (2025, June 26). *How automated security control assessment drive faster and smarter compliance audits*. CyberProof. https://www.cyberproof.com/blog/how-automated-security-control-assessment-drive-faster-and-smarter-compliance-audits/
- *What is Endpoint Security? EPP, EDR, and XDR explained*. (n.d.). Palo Alto Networks. https://www.paloaltonetworks.com/cyberpedia/what-is-endpoint-security
- DNS Tunneling: Ransomware Cybersecurity Definition | halcyon.ai. (n.d.). https://www.halcyon.ai/faqs/what-is-dns-tunneling
- Luna, C. D. (2024, September 9). *What is Industrial Control System (ICS) cyber security?* eSecurity Planet. https://www.esecurityplanet.com/cloud/industrial-control-systems-cyber-security/
- Command and Control, Tactic TA0011 - Enterprise | MITRE ATT&CK®. (n.d.-b). https://attack.mitre.org/tactics/TA0011/